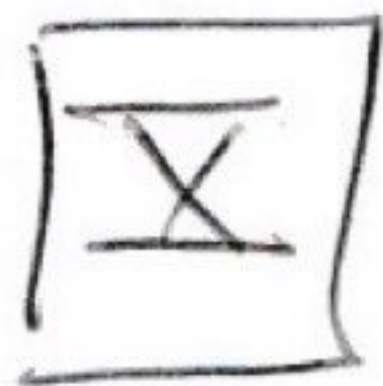


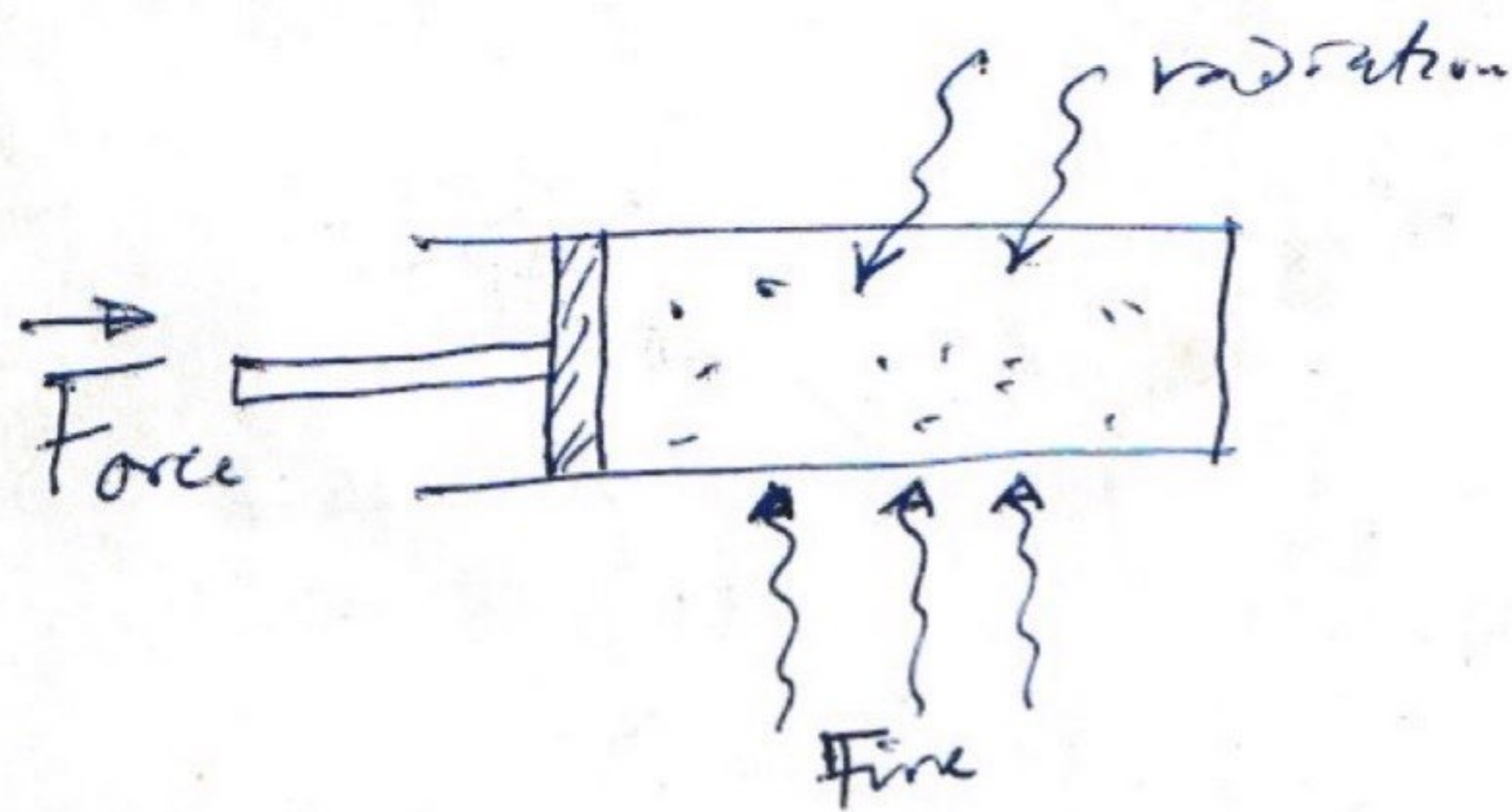
Heat



Heat is often considered as another form of energy. It is a manifestation of energy transfer by molecular collisions. It's a mode of energy exchange. Therefore, heat energy can be stored, transferred, or transformed from and converted into other forms of energy.

[In a solid and liquid, the atoms are in continuous motion but held together by attractive forces (set of springs). These motions result in a heating effect if energy is put into.]

Energy can be put into a substance in two ways. One way is to heat the substance. Another way is to perform some work on the substance. For example, air contained in a cylinder fitted with a piston can be given energy in two ways. Its internal energy may be increased by heat (Bunsen flame or radiation) or by work (force moving



piston). Its bulk energy is unchanged (macroscopic part of energy). It's composed of acceleration and potential energies.

A common unit for heat in scientific work is the Calorie (cal). It is the quantity of heat required to raise

All material things consist of atoms and molecules, and these are continually in motion, whether the object is gas, liquid or solid. Therefore even if ~~the~~ object is motionless, in spite of zero potential energy, it has an internal energy associated with the motion of atoms.

1 gram of water 1°C . With SI units, the Kilo Calorie is convenient. Its approximately the heat required to raise the temperature of 1 kg of water by 1K .

$$1\text{J} = 4,184 \text{ J/Kcal}$$

or

$$1\text{Cal} = 4186\text{J}$$

$$[1\text{ Joule of work} = 4.185\text{ Calories of heat}]$$

$$\text{Heat } Q = \frac{\text{Work}}{\text{Joule}}$$

Example

If a sample of water of 5 kg mass was brought from melting point (0°C) to room temperature (20°C), the number of Calories of heat supplied was

$$(5000)(20) = 100\,000\text{ Cal.}$$

This corresponds to an energy of

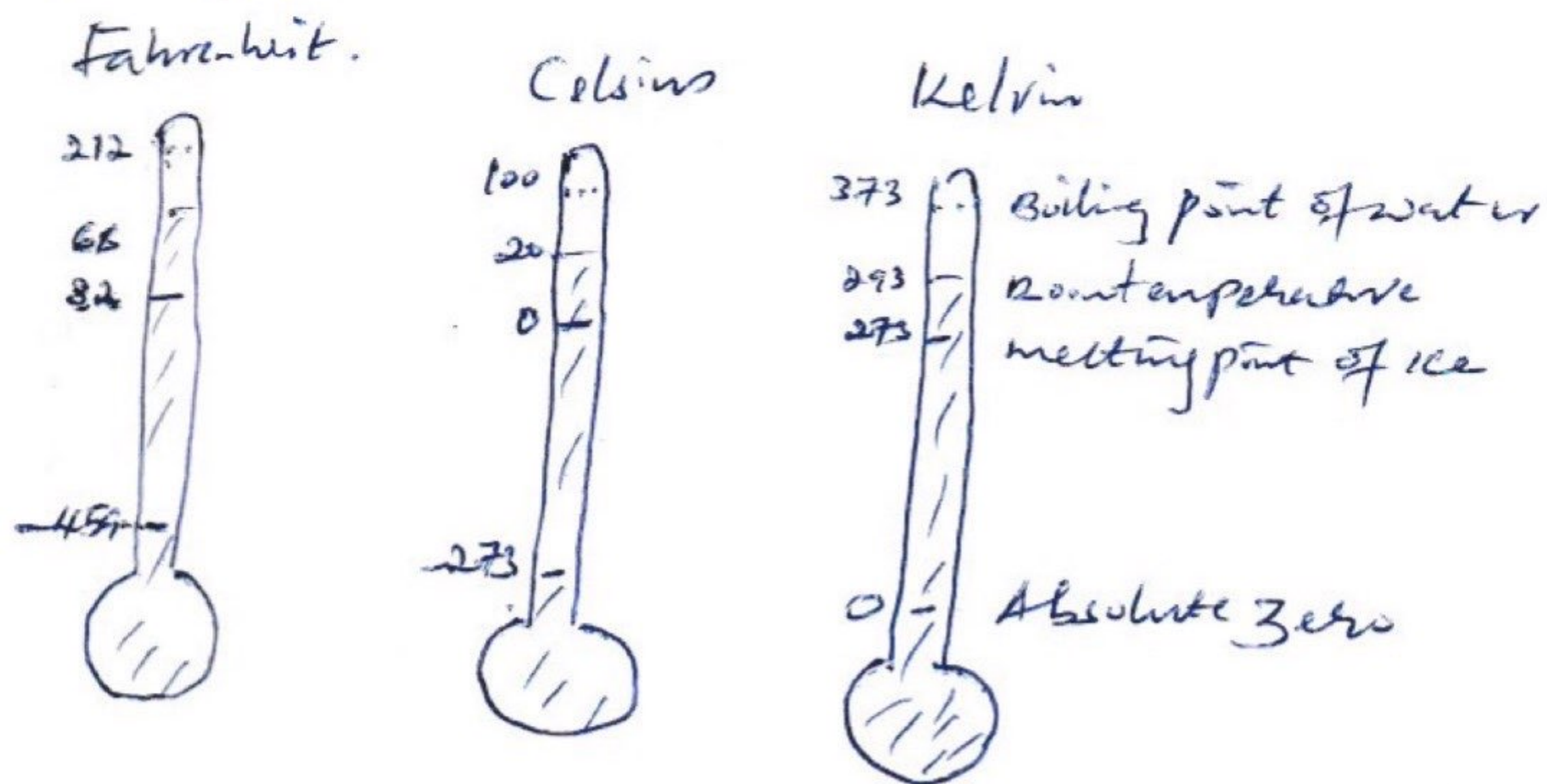
$$\frac{100000\text{ Cal}}{4.185\text{ Cal/J}} = 2.39 \times 10^3\text{ Joules}$$

Temperature Scales [hotness or coldness]

We shall treat temperature as a quantity that can be measured by a suitable device called a thermometer.

The most familiar type is the graduated glass tube with a coloured liquid, the level of which rises as the liquid expands and falls as it contracts. (Liquid thermometer).

The Celsius scale (C) is used in scientific measurements. The Kelvin Scale (K) or absolute scale uses the same spacing of degrees as the Celsius but its zero is at -273.15°C .



$$T(^{\circ}\text{F}) = \frac{9}{5} T(^{\circ}\text{C}) + 32 \quad \text{Celsius to Fahrenheit}$$

$$T(^{\circ}\text{C}) = \frac{5}{9} [T(^{\circ}\text{F}) - 32] \quad \text{Fahrenheit to Celsius}$$

Rankine scale

$$T(^{\circ}\text{R}) = \frac{9}{5} T(\text{K})$$

matter reacts to heat and cold in a variety of ways.

Eg When heated, a metal bar gets longer;

a liquid expands;

a gas expands, or its pressure increases, or both.

When very hot most substances glow and emit radiant energy. Certain marked changes may occur at definite temperatures — Freezing, boiling, igniting. In principle any of these effects could be used to define and measure temperature. For example an oven thermometer uses a metal strip. A typical outdoor thermometer uses a column of liquid. A device called a pyrometer measures high temperature by measuring emitted energy.

Temperature, by definition, is a scalar quantity. It has no directional property and no dependence on coordinates. Like ~~any~~ other definitions in physics, temperature is justified by experience. It is simple, self-consistent, and useful. A thermometer provides a single numerical value for T .

In general, any property of a substance exhibited when the substance gives ~~heat~~ energy can be used to quantify temperature.

Thermal expansion.

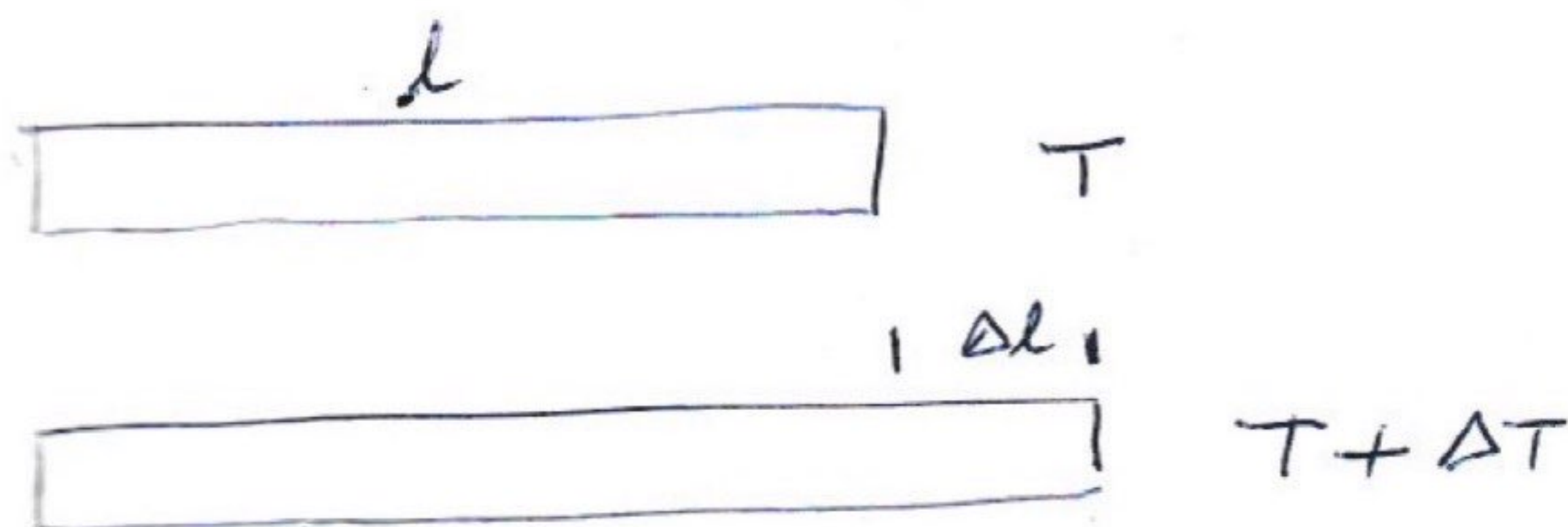
most substances increase in volume when heated, a process called thermal expansion. The addition of heat to any object tends to increase the thermal agitation of the atoms and molecules. On the average, the component particles are farther apart than at lower temperatures.

Materials expand about the same amount in each direction except in cases where the crystal structure is not the same along different axes.

A long thin bar of metal heated throughout from T to $T + \Delta T$ expands from length L to $L + \Delta L$. The length thus, increases by a fraction $\frac{\Delta L}{L}$. The linear coefficient of thermal expansion is defined by the ratio

$$\alpha = \frac{\Delta L/L}{\Delta T}$$

If α is known we can compute the change in length

$$\Delta L = L\alpha\Delta T$$


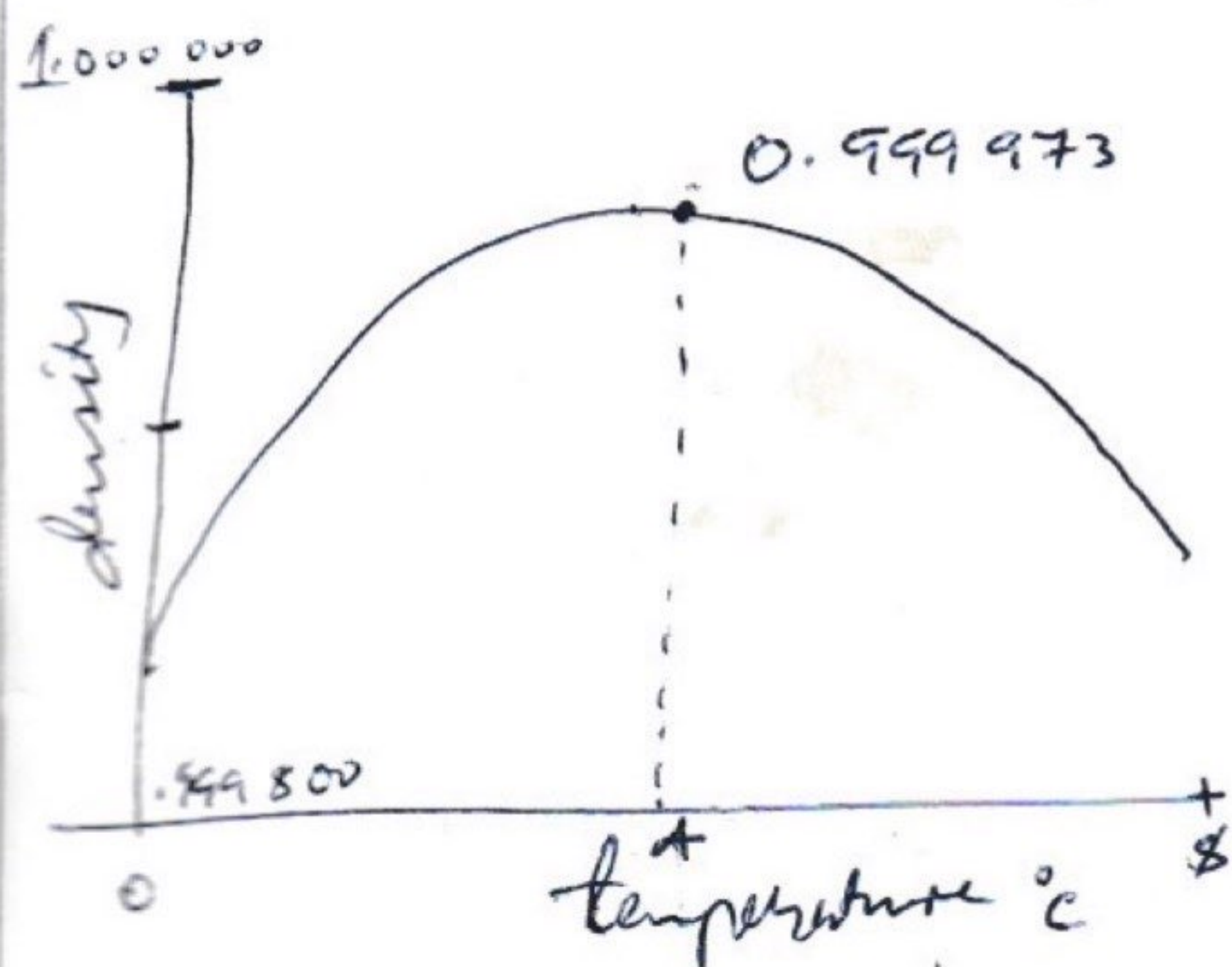
For example a thin wire of iron of length 0.5m is heated from 20°C to 100°C . Its coefficient α is about $11 \times 10^{-6}/^{\circ}\text{C}$. The increase in length is then

$$\Delta L = L \alpha \Delta T = (0.5\text{m})(11 \times 10^{-6}/^{\circ}\text{C})(80^{\circ}\text{C}) = 0.00044\text{m}.$$

Which is $\frac{1}{20}$ millimeter. Amounts of expansion depends on the type of material.

Mechanical thermometers, thermostats for controlling temperature are based on differences in expansion coefficients of metals. A bimetallic strip of Al and Cu.

Although the objects expand, the mass remain constant, ~~but~~ and density (mass per unit volume) decreases. Water,



expands and reaches its maximum density at 4°C . Below this temperature it increases in volume as shown therefore becoming less dense.

~~As~~ This anomalous expansion has a physical significance in that, cold water in winter would not be buoyed to the surface as it approaches freezing, but would sink to the bottom of bodies of water, which would soon be filled with ice. This would inhibit marine life.

Heat Capacity

The specific heat of a material is the amount of heat that must be added to a unit mass of the material to raise its temperature by 1 degree.

Symbol — C . Its mks units is $J/kg \cdot K$.

~~Specific~~ Specific heat depends on the material and the external conditions of heating. Two standard conditions are employed for defining specific heat: heating at constant volume and heating at constant pressure. The corresponding specific heats are written C_v and C_p . Specific heats of solids and liquids are usually specified at constant pressure. For gases, both C_v and C_p are of interest.

As noted before, 1 Cal of heat is required to raise

the temperature of 1 gm of water by 1 K. The specific heat of water is therefore

$$C_p(\text{water}) \cong 1 \frac{\text{Cal}}{\text{gm K}} = 1 \frac{\text{Kcal}}{\text{kg K}}$$

Water has a high specific heat compared to most materials.

Over the range from 0°C to 100°C, C for water differs by less than 1%.

From the definition of specific heat

$$\Delta Q = C m \Delta T$$

Which states that the amount of heat ΔQ required to raise the temperature of a mass m through a temperature interval ΔT is equal to the product of specific heat of the material, its mass and the temperature interval.

It's valid when C does not change significantly. ~~C~~

~~water~~

Example.

A 40 gm piece of Aluminum at 60°C is dropped into an insulated container of 100 gm of water at 15°C . If energy loss from the system of water plus aluminum is ignored, to what final temperature does the system come?

Sol.

The law of equilibrium tells us that both water and Al will come to the same temperature T_f . The law of conservation tells us that the heat gained by the water is equal to the heat lost by the aluminum or

$$\Delta Q(\text{H}_2\text{O}) + \Delta Q(\text{Al}) = 0.$$

$$\Leftrightarrow [C_p m \cdot (T_f - T_o)]_{\text{H}_2\text{O}} + [C_p m \cdot (T_f - T_o)]_{\text{Al}} = 0.$$

$$\Rightarrow 1.00 \times 100 \times (T_f - 15) + 0.22 \times 40 \times (T_f - 60) = 0$$

C_p is expressed in $\text{cal/gm}^{\circ}\text{K}$, m is expressed in gm, and T is expressed in $^{\circ}\text{C}$. The solution for T_f is

$$T_f = 18.6^{\circ}\text{C}$$

After this H/W # No 10.

Change of state.

A phase change is a structural change in a substance that occurs at a specific temperature and pressure. Phase changes include vaporization, fusion, sublimation and changes from one crystalline form to another. If a change of internal energy (and, with it, a change of density) accompanies the change of phase it is called a 1st order phase transition. Melting and boiling are 1st order transitions.

The energy change per unit mass associated with fusion, or its opposite melting, is called heat of fusion.

The energy change per unit mass associated with vaporization is called heat of vaporization. For water at normal pressure,

$$l_{\text{fusion}}(\text{H}_2\text{O}) = 80 \text{ cal/gm (or kcal/kg)}$$

$$l_{\text{vaporization}}(\text{H}_2\text{O}) = 539 \text{ cal/gm (or kcal/kg)}$$

These quantities are also known as latent heats

Q2 Tut 1

Example.

To 1kg of ice at 0°C and pressure of 1 atm is added 240 kcal of heat. What is the fate of ice? melting requires 80 kcal then raising it; then raising its temperature to 100°C requires another 100 kcal. This leaves 60 kcal, enough to vaporize only $\frac{1}{9}$ of the kilogram. The ice is therefore converted to a mixture of 0.89 kg of water and 0.11 kg of steam both at 100°C .

Heat transfer. Conduction, convection, radiation.

The energy of thermal motion is found to be transferred in solid and liquid bodies according to the same rule as for gases

$$H = -K \frac{\Delta T}{\Delta x}$$

Where K is the thermal conductivity

In a time t , through a ^{solid} material of crosssectional area A , the total amount of heat transferred is given by

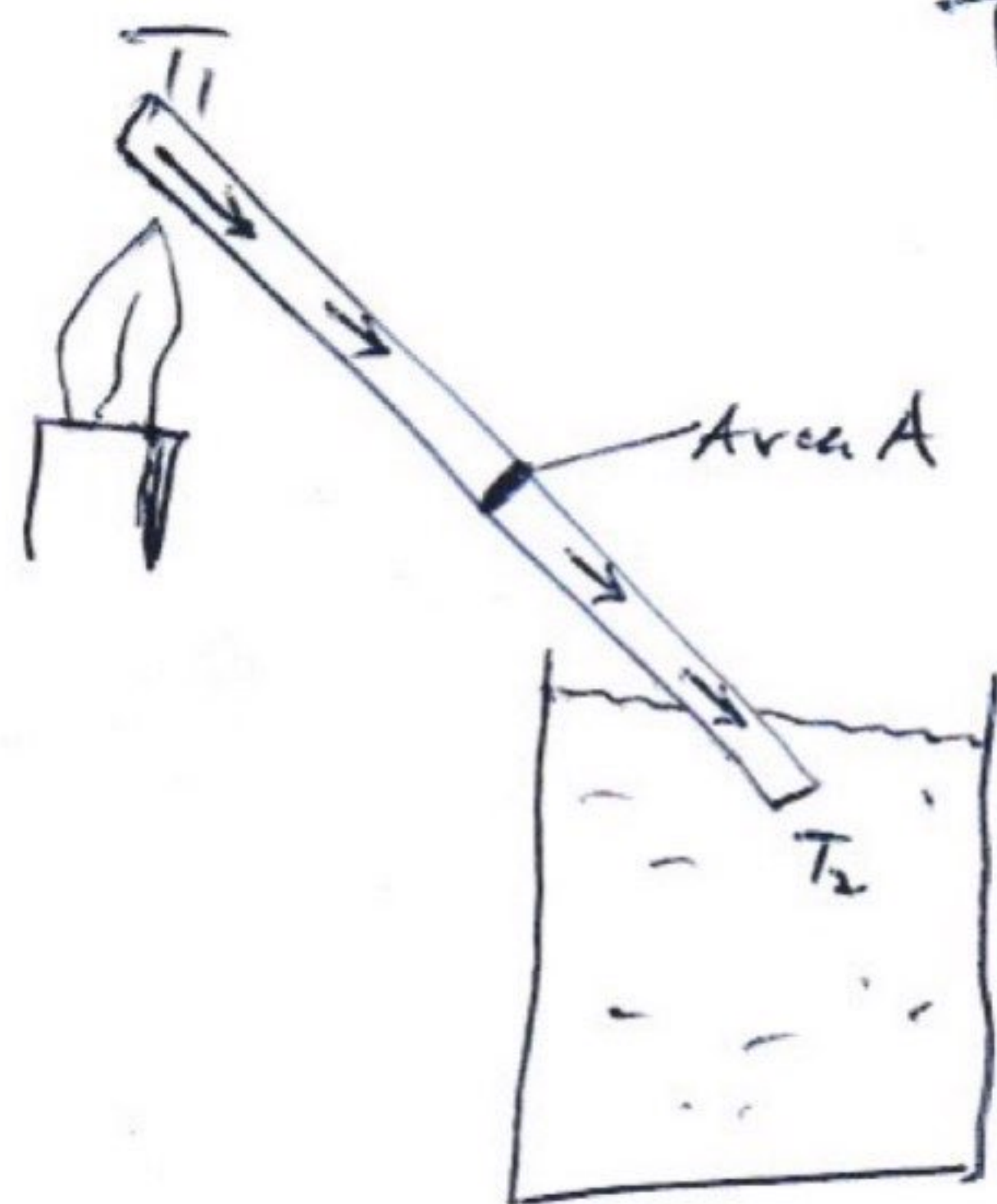
$$Q = \frac{K A (T_2 - T_1) t}{L}$$

L = length

A = area

t = time s

$T_2 - T_1$ = temperature change.



The conduction of heat is poor in liquids and gases in which the mechanism of convection is responsible for most of the heat transfer. This process involves motion of large

Amount of fluid due to temperature differences and buoyancy.
Natural or free convection is a circulation that takes place automatically when a fluid is in contact with a hot surface.
Forced convection involves a flow of fluid over the surface that continually sweeps away heated particles.

Hot air rises and cold air settles in a room. This is due to the lower density of the heated gas which causes it to be buoyed up. Processes which transfer heat energy ~~can~~ be explained from a molecular point of view. Molecules of air strike the heated surface, gain energy at the expense of the surface, and move on with the stream.

Some properties of the cooling agent and boundary are important in increasing the rate of heat transfer.

high velocity (heated fluid replaced quickly)

large specific heat

large conductivity (ability to transfer heat)

high degree of internal motion (much exchange of fluid)

large surface area

large difference in temperature between surface and fluid

When the surface ~~is~~ is raised to very high T compared to the surroundings, heat is transferred by radiation. This radiation is in the infrared light. (If frequency lower than

the visible range) The skin of the body will detect these rays through a warming effect. By use of sensitive heat detectors, infrared photos can be made in complete darkness. The heat from the sun comes to us by these rays.